

# Ethan Arnold NE 533 Exam 1

1.  $\epsilon = 19\%$   $\rho = 14.3 \text{ g/cm}^3$   $\sigma = 587 \text{ b} = 587 \times 10^{-24} \text{ cm}^2$   
 $n = 4 \times 10^{12} \frac{\text{A}}{\text{cm}^2 \text{ s}}$   $p_{\text{VO}_2} = 10.97 \text{ g/cc}$

$Q = E_f \times N_f \times \sigma_f \times d$ ,  $M_{\text{VN}} = (0.19(235) + 0.81(238)) + 14$

$M_{\text{VN}} = 251.43 \frac{\text{g}}{\text{mol}}$

$N_f = 14.3 \text{ g/cm}^3 \times \frac{1}{251.43 \text{ g/mol}} \times \frac{6.022 \times 10^{23} \text{ atoms}}{1 \text{ mol}} \times \frac{1 \text{ V}}{1 \text{ V}} \times 0.19$

$N_f = 6.507 \times 10^{21} \frac{\text{atoms}}{\text{cm}^3}$

$Q = (200 \times 10^6 \text{ eV}) (1.602 \times 10^{-19} \frac{\text{J}}{\text{eV}}) (6.507 \times 10^{21} \frac{\text{atoms}}{\text{cm}^3}) (587 \times 10^{-24} \text{ cm}^2) (4 \times 10^{12} \frac{\text{A}}{\text{cm}^2 \text{ s}})$

$Q = 489.56 \frac{\text{W}}{\text{cm}^3}$

$N_f = 6.507 \times 10^{21} \frac{\text{atoms}}{\text{cm}^3} = 10.97 \text{ g/cm}^3 \times \frac{1}{M_{\text{VO}_2}} \times \frac{6.022 \times 10^{23} \text{ atoms}}{1 \text{ mol}} \times \frac{1 \text{ V}}{1 \text{ V}} \times (x)$

$M_{\text{VO}_2} = (x(235) + (1-x)238) + 2(16)$

$6.507 \times 10^{21} \frac{\text{atoms}}{\text{cm}^3} = 10.97 \frac{\text{g}}{\text{cm}^3} \times \frac{1}{(x(235) + (1-x)238) + 32} \times \frac{6.022 \times 10^{23} \text{ atoms}}{1 \text{ mol}} \times \frac{1 \text{ V}}{1 \text{ V}} \times (x)$

$x = 26.5\% \text{ enrichment}$

2.  $k_c = 0.18 \frac{\text{W}}{\text{cm K}}$ ,  $k_f = 0.04 \frac{\text{W}}{\text{cm K}}$   $h = 1.5 \frac{\text{W}}{\text{cm}^2 \text{ K}}$ ,  $Q = 350 \frac{\text{W}}{\text{cm}^3}$ ,  $R_f = 0.45 \text{ cm}$   
 $t_g = 80 \mu\text{m}$ ,  $t_c = 0.06 \text{ cm}$   $T_{\text{cool}} = 540 \text{ K}$   $t_{c,0.5} \text{ and } r = 0.1$

outside-in:  $T_{\text{oc}} = \frac{Q}{2h_{\text{cool}}} R_f + T_{\text{cool}} \rightarrow T_{\text{oc}} = \frac{350 \frac{\text{W}}{\text{cm}^3}}{2(1.5 \frac{\text{W}}{\text{cm}^2 \text{ K}})} (0.45 \text{ cm}) + 540 \text{ K}$

$T_{\text{oc}} = 592.5 \text{ K}$

$T_{\text{ic}} = \frac{Q}{2h_{\text{clad}}} R_f + T_{\text{oc}}$ ,  $h_{\text{clad}} = \frac{0.18 \frac{\text{W}}{\text{cm K}}}{0.06 \text{ cm}} \rightarrow h_{\text{clad}} = 3 \frac{\text{W}}{\text{cm}^2 \text{ K}}$

$h_c = 3 \frac{\text{W}}{\text{cm}^2 \text{ K}}$ ,  $T_{\text{ic}} = \frac{350 \frac{\text{W}}{\text{cm}^3}}{2(3 \frac{\text{W}}{\text{cm}^2 \text{ K}})} (0.45 \text{ cm}) + 592.5 \text{ K} \rightarrow T_{\text{ic}} = 618.75 \text{ K}$

Temp profile in clad is linear so  $T_{\text{c,mid}} = \frac{T_{\text{ic}} + T_{\text{oc}}}{2} \rightarrow T_{\text{c,mid}} = \frac{592.5 \text{ K} + 618.75 \text{ K}}{2}$

$T_{\text{c,mid}} = 605.63 \text{ K}$



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2. (cont.)

$$T_s = \frac{Q}{2 \ln r_{gap} R_f + \frac{\pi}{k_{gap}}} \quad h_{gap} = \frac{k_{gap}}{r_{gap}}; \quad k_{gap} = 16 \times 10^{-6} \times (616.75 K)^{0.79}$$

$$k_{gap} = 0.0026 \frac{W}{cm K} \quad \ln r_{gap} = \frac{0.0026 \frac{W}{cm K}}{0.008 cm} \rightarrow \ln r_{gap} = 0.321 \frac{W}{cm^2 K}$$

$$T_s = \frac{350 \frac{W}{cm^3}}{2 (0.321 \frac{W}{cm^2 K})} (0.45 cm) + 616.75 K \rightarrow T_s = 864.08 K$$

$$T(r=0.1 cm) = \frac{Q (R_f^2 - r^2)}{4 k} + T_s, \quad k = 0.04 \frac{W}{cm K}$$

$$T(r=0.1 cm) = 350 \frac{W}{cm^3} \frac{((0.45 cm)^2 - (0.1 cm)^2)}{4 (0.04 \frac{W}{cm K})} + 864.08 K$$

$$T(r=0.1 cm) = 1285.2 K$$

3.  $L = 3.6 m$ ,  $LHR^0 = 400 \frac{W}{cm}$ ,  $\gamma = 1.25$ ,  $LHR(2.1 m)$ ,  $T_0$

$\rho = 4200 J/kg K$ ,  $\dot{m} = 0.1 kg/s-rad$ ,  $T^{\text{in}} = 580 K$

$$Z_0 = \frac{L}{2} = \frac{3.6 m}{2} = 1.8 m$$

$$LHR\left(\frac{2.1 m}{1.8 m}\right) = LHR^0 \cos\left[\frac{\pi}{2\gamma}\left(\frac{Z}{Z_0} - 1\right)\right]$$

$$LHR\left(\frac{2.1 m}{1.8 m}\right) = 400 \frac{W}{cm} \cos\left[\frac{\pi}{(2)(1.25)}\left(\frac{2.1}{1.8} - 1\right)\right] \rightarrow LHR\left(\frac{2.1}{1.8}\right) = 391.26 \frac{W}{cm}$$

$$T_{cool}(3.6) - T^{\text{in}}_{cool} = \frac{2\gamma}{\pi} \frac{Z_0 LHR^0}{\dot{m} \rho} \left\{ \sin\left(\frac{\pi}{2\gamma}\right) + \sin\left[\frac{\pi}{2\gamma}\left(\frac{Z}{Z_0} - 1\right)\right] \right\}$$

$$T_{cool}(3.6) = \frac{2(1.25)}{\pi} \frac{(1.8 m)(400 \frac{W}{cm})}{(0.1 kg/s)(4200 J/kg K)} \left\{ \sin\left(\frac{\pi}{2(1.25)}\right) + \sin\left[\frac{\pi}{2(1.25)}\left(\frac{3.6}{1.8} - 1\right)\right] \right\} + 580 K$$

$$T_{cool}(3.6) = 639.48 K$$

4.  $FIMA = 0.05$ ,  $T_{avg} = 1200 K$ ,  $A = 3.8 + 200 FIMA$ ,  $B = 0.0217 cm/W$

OFIMA:  $A = 3.8 + 200(0) = 3.8 \frac{cm K}{W}$ ,  $k_{ox} = \frac{1}{(3.8 \frac{cm K}{W} + 0.0217 cm/W (1200 K))}$

$$k_{ox} = 0.0335 \frac{W}{cm K}$$

50% FIMA:  $A = 3.8 + 200(0.05) = 13.8 \frac{cm K}{W}$ ,  $k_{ox} = \frac{1}{(13.8 \frac{cm K}{W} + 0.0217 cm/W (1200 K))}$

$$k_{ox} = 0.0251 \frac{W}{cm K}$$

$$\% \text{ degradation} = \frac{0.0251 - 0.0335}{0.0335} (100) = 25.1\% \text{ decrease}$$



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5. Steady state, constant thermal conductivity,  $z$  independent of  $T$ , and axisymmetric.

6.  $U_3O_8$  is turned into gaseous  $UF_6$ . The  $UF_6$  gas travels into a centrifuge which rapidly spins. The heavier  $U-238$  is pushed out to the outside of the centrifuge by centrifugal force while the lighter  $U-235$   $UF_6$  is removed from the central region of the centrifuge.

7. The DNBR is the ratio that describes how close a rod is to the critical heat flux.  $DNBR = \frac{CHF}{Q_{rod}}$ . As the ratio approaches 1, the rod is approaching the critical heat flux. If the critical heat flux is reached, the rod surface can transition from nucleate boiling (some bubbles on rod surface due to  $T > T_{sat}$ ), to transition or film boiling where vapor layers appear on the rod surface and dryout can occur. This can cause the coolant to no longer cool the rod as effectively leading to hotspots and rod failure. Most fuels groups have DNBR's  $\approx 1.15$  to  $1.3$  to protect against uncertainty in DNB calculations and accidents.

8. Swear density is given by the pellet volume divided by the rod volume and reduces to  $\frac{r_p^2}{r_c^2}$ . Swear density accounts for the pellet-clad gap. The pellet-clad gap allows the fuel to swell with irradiation/temperature and prevents strain failures.

9. fission gas release (porosity), fuel fragmentation

10. Cladding separates fuel from the coolant, prevents fuel corrosion by the coolant, maintains fission product release, and keeps the pellets in place. Zr is a good cladding as it is a low neutron absorber,

has good thermal conductivity, is widely available, and does not chemically interact with  $UO_2$ .



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4/4 11. fuel, clad, gap, coolant

12. How much heat does it transfer to the coolant?

How long can it operate without a problem?

How does it perform under accident conditions?

13. Finite difference - strength: relatively easy, weakness: poor ability to handle complex geometries

Finite volume - strength: can handle more diverse geometries, weakness: can struggle with complex boundary conditions

Finite element - can handle all geom's, BC's, IC's, weakness: very complex to implement and can be computationally expensive

14. positive: high melting temp, relatively radiation resistant and stable in water  
negative: brittle, low linear heat rate

15. Implicit time integration calculates the next step using conditions from a point at that future step (e.g., backwards Euler method uses  $y_{n+1} = y_n + dt y'_{n+1}$ ). Explicit time integration calculates the next step using conditions from a present point in time (e.g., forward Euler method uses  $y_{n+1} = y_n + dt y'_n$ ).  
Implicit is more complex but can be more stable than explicit for certain discretizations.